

Digital Forensic Investigation Report using Python

Python Program:

```
import datetime

def generate_forensic_report(case_id, findings, investigator):
    lines = []
    lines.append("DIGITAL FORENSIC INVESTIGATION REPORT")
    lines.append(f"Case ID: {case_id}")
    lines.append(f"Investigator: {investigator}")
    lines.append(f"Report generated: {datetime.datetime.now()}")
    lines.append("FINDINGS:")
    for i, finding in enumerate(findings, 1):
        lines.append(f"{i}. {finding}")
    lines.append("CONCLUSION:")
    lines.append("Evidence indicates unauthorized access consistent with a brute-force attack.")
    return "\n".join(lines)

findings = [
    "203.0.113.99 attempted 5 failed logins within 20 seconds.",
    "SHA-256 hash verified unchanged.",
    "Email header traced to a spoofed domain."
]

report = generate_forensic_report("CASE-2026-014", findings, "Analyst B")
print(report)
with open("forensic_report.txt", "w") as f:
    f.write(report)
```

Sample Output:

```
DIGITAL FORENSIC INVESTIGATION REPORT
Case ID: CASE-2026-014
Investigator: Analyst B
Report generated: 2026-07-23 09:18:33
FINDINGS:
1. 203.0.113.99 attempted 5 failed logins within 20 seconds.
2. SHA-256 hash of evidence file verified unchanged.
3. Email header traced to a spoofed domain.
CONCLUSION: Evidence indicates unauthorized access consistent with a brute-force attack.
Report saved successfully as forensic_report.txt
```

Explanation:

This program demonstrates how to generate a simple digital forensic investigation report. It records the case ID, investigator name, report generation time, forensic findings, and conclusion. The completed report is displayed on the screen and also saved as a text file. This illustrates the reporting stage of a digital forensic investigation and provides a structured report suitable for documentation and legal review.